

Total No. of Questions : 8]

SEAT No. :

PE-4230

[Total No. of Pages : 2

[6582]-1

S.E. (Civil)

GEOTECHNICAL ENGINEERING
(2019 Pattern) (Semester - IV) (201008)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8
- 2) Neat diagrams must be drawn whenever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data if necessary and mention it clearly.
- 5) Use of Calculator is allowed.

- Q1)** a) Define Compaction. Compare between Compaction and consolidation. [6]
b) A soil sample has specific gravity 2.62 and Maximum Dry Density 1650 kg/m³ at a moisture content of 10%. Find Degree of Saturation, air content and percentage air voids at Maximum Dry density. [6]
c) What is pressure bulb? Explain its significance and draw neat sketch of pressure bulb for concentrated point loading. [5]

OR

- Q2)** a) Write the equation for stresses in soil for point loading by Boussinesq's and Westergaard's theory and assumptions of these theories. [5]
b) A concentrated load of 450 kN is applied at a ground surface compute the vertical pressure [6]
i) At a depth 6m below the load
ii) At a distance 6m at same depth.
c) Explain standard proctor compaction test. [6]

- Q3)** a) Explain in detail Skempton's Pore pressure parameters. [5]
b) A soil sample fails under an axial stress of 250 kN/m² when laterally unconfined. The failure wedge is making an angle of 50° with horizontal. Calculate Cohesion & angle of internal friction of soil. [6]
c) Write short note on Thixotropy and Sensitivity of clay. [6]

OR

P.T.O.

- Q4)** a) Explain in detail factors affecting shear strength of soil. [5]
 b) In vane shear test on clay, the following observations are made Applied Torque : 180 kg cm, Height of vane : 10cm Diameter of Vane : 5cm, calculate shear strength of clay. [6]
 c) Explain in detail Triaxial Compression Test. [6]

- Q5)** a) Explain in detail earth pressure at rest. Draw pressure distribution diagram for retaining wall with height h and Unit weight of soil in backfill $= \gamma$. [6]
 b) Write short note on Coulomb's Wedge Theory in detail. [6]
 c) A smooth vertical wall retains a level backfill with $\gamma = 20 \text{ kN/m}^3$, $\Phi = 32^\circ$ and $C=0$ to a depth of 8m. Draw lateral pressure diagram and compute the total thrust on the retaining wall. What will be active pressure if water stands at a depth of 3.5m? [6]

OR

- Q6)** a) Explain Rebhann's graphical method for determination of earth pressure on retaining wall. [6]
 b) In a cohesionless soil deposit having unit weight of 20 kN/m^3 and $\Phi = 32^\circ$. Determine active and passive lateral pressure intensities at depth of 8m. [6]
 c) Discuss points of difference between Rankines and Coulombs theory of earth pressure. [6]
- Q7)** a) Determine the critical height of excavation for vertical cut in cohesive soil, if $C=20\text{kN/m}^3$ and $\gamma = 20\text{kN/m}^3$. [6]
 b) Differentiate between Finite and Infinite slopes. [6]
 c) What is stability number? What is its utility in the analysis of stability of slopes. [6]

OR

- Q8)** a) Determine the stability number for vertical cut in a cohesive soil with following details, $\gamma = 18 \text{ kN/m}^3$, $C = 15 \text{ kN/m}^3$, $\Phi = 32^\circ$. [6]
 b) What are the assumptions that are generally made in the analysis of slopes? Discuss validity of such assumptions in brief. [6]
 c) Discuss various types of slope failures in detail. [6]

